

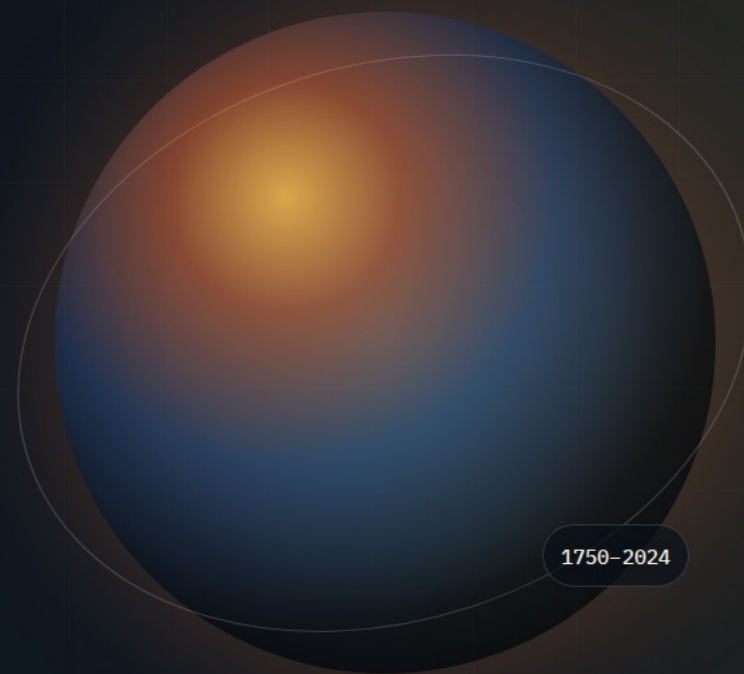
Global CO₂ Emissions

Trends, regional shifts, and per-capita inequalities from the Industrial Revolution to 2024.

Scope: 1750–2024

Audience: climate negotiators

Lens: totals + equity



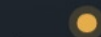
ATMOSPHERIC CONCENTRATION

CO₂ reaches **429 ppm** — over 50% above pre-industrial levels

NOAA's Mauna Loa Observatory recorded 429 ppm in February 2026, compared with a pre-industrial baseline of approximately 280 ppm.

429

PARTS PER MILLION · FEBRUARY 2026



~280 ppm
Pre-industrial



1958
Mauna Loa record begins



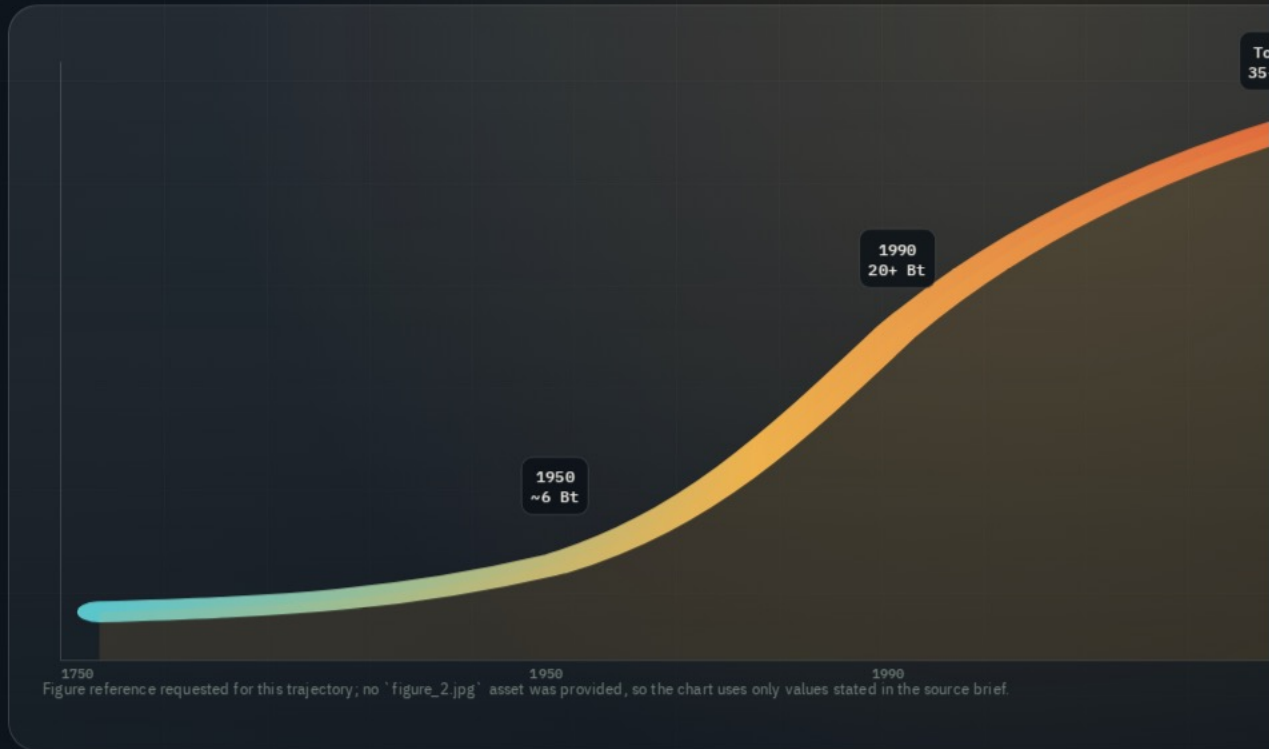
429 ppm
Feb 2026

The current level is at least 1.5× greater than the natural increase at the end of the last ice age; that prior shift unfolded over thousands of years, while this change occurred in ro centuries.

GLOBAL EMISSIONS TRAJECTORY

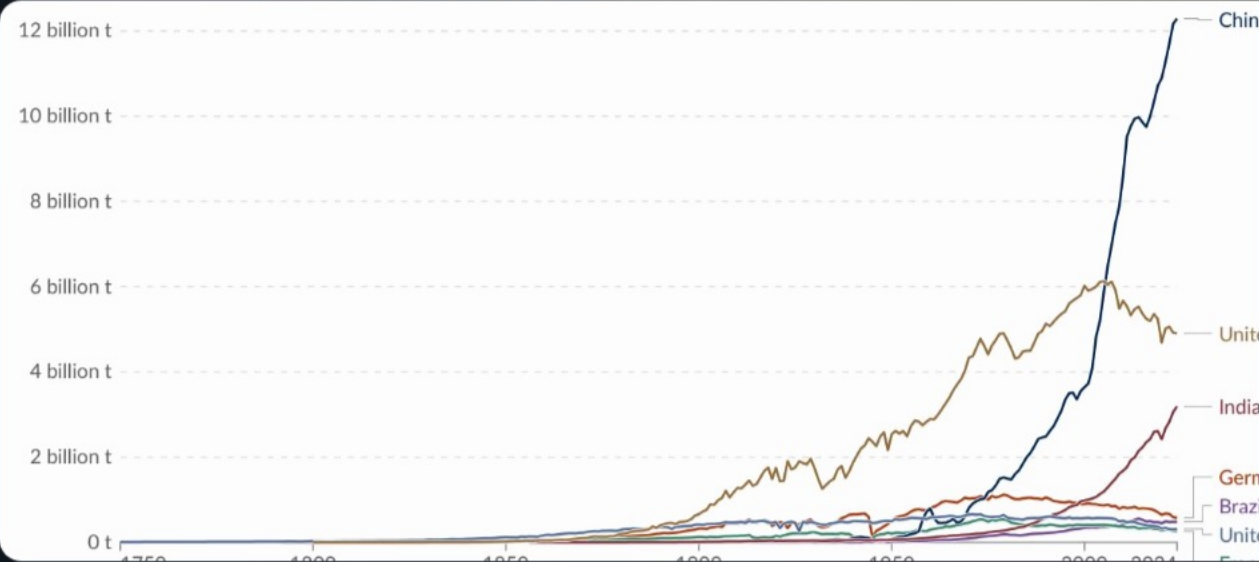
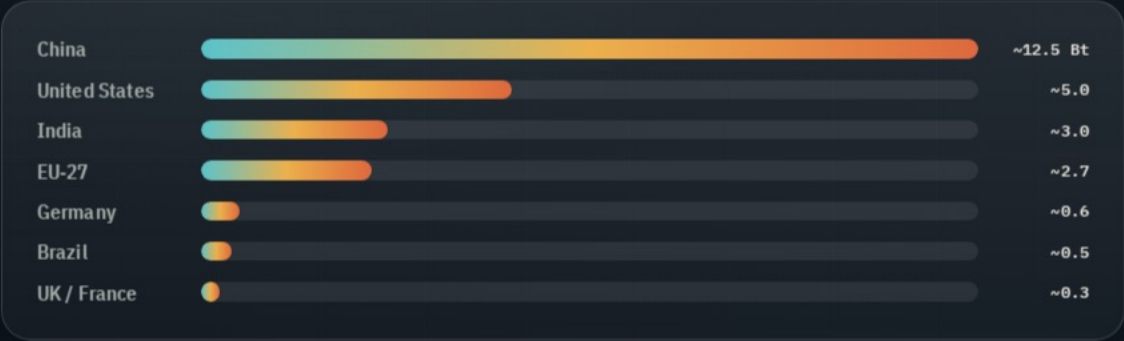
Fossil fuel CO₂ grew
from **6 Bt** in 1950 to **35+**
Bt today

Emissions were negligible before the Industrial Revolution, grew slowly until the mid-20th century, nearly quadrupled by 1990, and have not yet peaked.



COUNTRY-LEVEL TOTALS

China emits **~12.5 Bt**—
more than double the
United States



Reference: figure_3.jpg. China accounts for ~27% of global emissions; the US is ~14% and declining from its ~6 Bt peak around 2005.

REGIONAL SHARE SHIFT

Asia now produces
~50%; US + Europe are
below one-third

1900

Europe and the US accounted for **more than 90%** of global emissions.



- Industrial-era concentration in Europe + US
- All other regions: small remainder

2024

Asia produces roughly **half**; US + Europe are **below one-third**.



- Asia: roughly half of all emissions
- US + Europe: below one-third combined
- Africa: 3-4%; South America: 3-4%

The visual is a side-by-side comparison of known source shares; it communicates the dramatic re-centering of annual emissions from historic industrial economies toward Asia.

PER-CAPITA INEQUALITIES

Per capita gap: **15 t** in the US vs. **0.1 t** in Chad

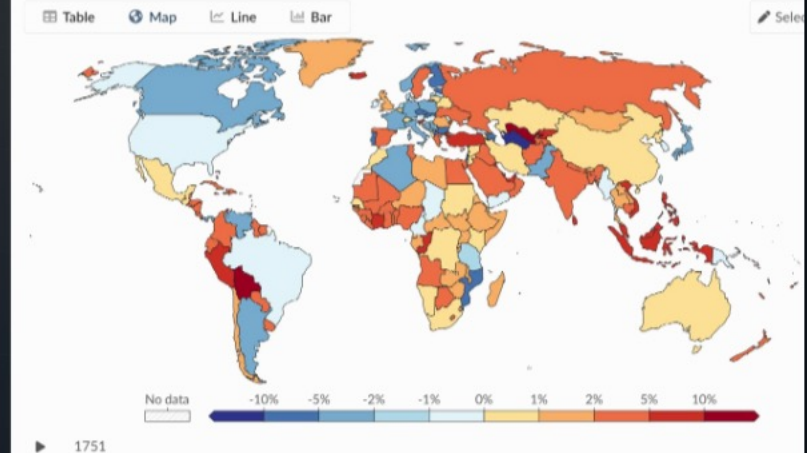


150×

The average American or Australian emits in under two days what the average person in Chad or Niger emits in an entire year.

Annual percentage change in CO₂ emissions, 2024

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.



Data source: Global Carbon Budget (2025) - [Learn more about this data](#)

Download Share Enter

Reference: figure_1.jpg

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

Our World
in Data

Table

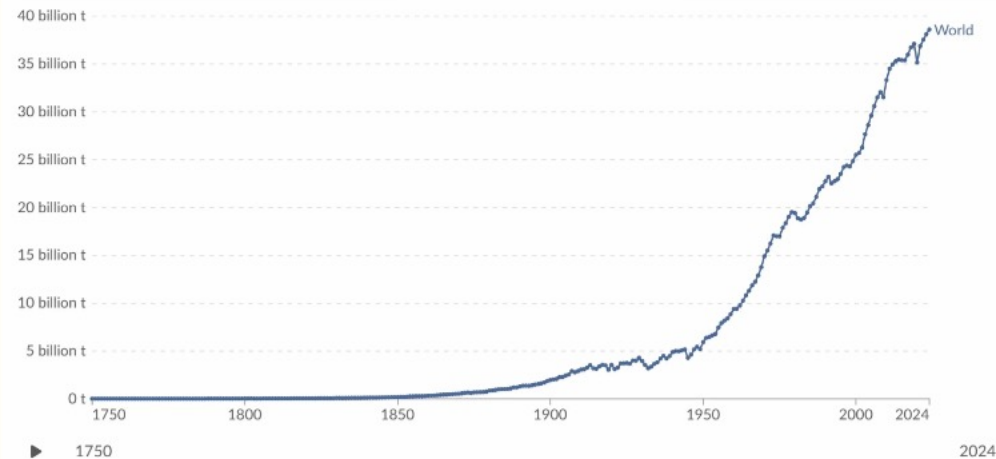
Map

Line

Bar

Edit countries and regions

Settings



Data source: Global Carbon Budget (2025) - [Learn more about this data](#)
OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

Download

Share

Enter full-screen

ANNUAL CHANGE SNAPSHOT

2024 shows a mixed map of cuts and increases

Declines: the United States and much of Europe recorded roughly -1% to -5%, shown in blue.

Increases: Russia, parts of Southeast Asia, and several African nations rose by +2% to +10% or more.

Outliers: one or two South American countries experienced very large percentage increases above +10%, while others declined.

Reference: figure_4.jpg — annual percentage change in CO₂ emissions, 2024.

ENERGY MIX AND POLICY CHOICES

Similar prosperity can yield very different per-capita CO₂

France and the UK have much lower per capita emissions than Germany or the Netherlands, largely because of their energy mix: higher nuclear and renewable shares versus higher fossil fuel reliance.

COUNTRY GROUP	ENERGY SOURCES HIGHLIGHTED	PER CAPITA CO ₂
France / UK	Higher nuclear and renewable shares; declining emissions	France ~4 t; UK ~4
Germany / Netherlands	Higher fossil fuel share than France / UK	Higher than France
Source brief provides explicit per-capita values for France and the UK and a directional contrast for Germany / Netherlands, no unsupported exact values are added.		

SYNTHESIS FOR NEGOTIATORS

Emissions are still rising, but the map of responsibility is shifting

35+ Bt

Global annual fossil fuel and industry CO₂ emissions exceed 35 billion tonnes and have not yet peaked.

>25%

China alone accounts for over one-quarter of global emissions, at ~12.5 Bt annually.

~50%

Asia now produces roughly half of emissions; US + Europe are below one-third combined.

150×

Per-capita gaps remain extreme, from ~15 t in the US / Australia to ~0.1 t in Chad / Niger.

Mix

Energy mix and policy choices can materialize per-capita outcomes among wealthy economies.